

MATHEMATICAL POSTULATES OF QUANTUM MECHANICS

Unit 2: Linear Algebra

Complex vectors and matrices, the inner product and what it measures, and the special matrices that quantum mechanics is written in.

Prepared by **Dr. Abhijith Anandakrishnan**

Assistant Professor, Amrita School of AI

COURSE

25AID432 Mathematical Postulates of
Quantum Mechanics

UNIT

2 of 3

SESSIONS

4 to 9

PREREQUISITE

Unit 1, complex numbers

READING TIME

About two and a half hours

OFFERING

Odd Semester 2026-27

Before you start

Unit 1 gave you complex numbers. This unit gives you the two objects that quantum mechanics is built from: **vectors** and **matrices**.

Here is the whole idea in one page, with no physics detail. A quantum system, such as a single qubit, is described at any moment by a list of complex numbers. That list is called its **state**. For one qubit the list has two entries; for two qubits it has four; for ten qubits it has 1024. Mathematically, such a list is a **vector**.

Anything you do to the system (a gate in a quantum computer, a pulse of light on an atom, simply waiting) changes that list of numbers into another list, and it does so in a very particular way: each new entry is a fixed weighted sum of the old entries. A rule of that kind is exactly what a **matrix** does. So operations are matrices, and applying an operation is multiplying a matrix by a vector.

Measurement asks a question of the form "how much of this state points along that direction?" The tool that answers it is the **inner product**, and it produces the probabilities you will meet in Unit 3.

That is the plan of the unit:

YOU WILL LEARN	BECAUSE IN UNIT 3 IT BECOMES
Complex vectors	Quantum states (kets)
Matrices acting on vectors	Quantum operations (gates)
Inner product and norm	Overlaps and probabilities
Orthonormal bases and projection	Measurement in a basis
Hermitian and unitary matrices	Observables and allowed operations
Eigenvalues, tensor product	Measurement outcomes, several qubits

None of this needs calculus. It needs arithmetic with complex numbers, done carefully, and a willingness to keep track of which entry is which. Most mistakes in this unit are bookkeeping mistakes, and the notes name them as they come up.

BY THE END OF THIS UNIT YOU WILL BE ABLE TO

- Add complex vectors, multiply them by complex scalars, and form linear combinations.
- Compute the transpose and the conjugate transpose (dagger) of vectors and matrices, including the dagger of a product.
- Multiply matrices by vectors and by matrices, say when a product is defined, and show that the order of multiplication matters.
- Invert a 2×2 matrix and check the answer.
- Compute inner products and norms of complex vectors with the physics convention, normalise a vector, and test two vectors for orthogonality.
- Check whether a set of vectors is an orthonormal basis, and expand any vector in it using inner products.
- Recognise diagonal, symmetric, Hermitian and unitary matrices, and state what each property guarantees.
- Find the eigenvalues and eigenvectors of a 2×2 matrix, and compute tensor products of vectors and matrices.
- Check any of these hand calculations in NumPy.

KEY TERMS IN THIS UNIT

vector · entry · scalar · zero vector · linear combination · transpose · conjugate transpose (dagger) · row vector · column vector · matrix · shape · matrix product · non-commutativity · identity matrix · inverse · determinant · inner product · norm · unit vector · normalisation · orthogonal · linear

independence · span · basis · standard basis · orthonormal basis · component · projection · diagonal matrix · symmetric matrix · Hermitian matrix · unitary matrix · eigenvalue · eigenvector · characteristic equation · tensor product · Kronecker product

Vectors

Session 4.

Column vectors in $\mathbb{R}^n$

A **vector** is an ordered list of numbers, written as a column. The numbers are its **entries**, numbered from 1:

$$\vec{v} = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix}.$$

If all the entries are real, the vector lives in $\mathbb{R}^n$, the set of all columns of n real numbers. With $n = 2$ you can draw it: the vector $\begin{pmatrix} 3 \\ 1 \end{pmatrix}$ is an arrow from the origin to the point $(3, 1)$ of the plane. With $n = 3$ you can still picture it. From $n = 4$ onwards you cannot, and it does not matter: every rule in this unit is stated through the entries, and the entries do not care how many of them there are.

Order matters. $\begin{pmatrix} 3 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 3 \end{pmatrix}$ are different vectors.

NOTE · WRITING COLUMNS INSIDE A SENTENCE

A column takes a lot of vertical space. When a vector appears in running text, these notes write its entries on one line with commas, such as $\vec{v} = (3, 1)$. That still means the **column**. A genuine row is always written with matrix brackets and no commas, such as $(3 \ 1)$.

Complex vectors: $\mathbb{C}^n$

Quantum mechanics needs complex entries. The set of all columns of n complex numbers is written $\mathbb{C}^n$. For example,

$$\vec{u} = \begin{pmatrix} 1+i \\ 2 \end{pmatrix} \in \mathbb{C}^2, \quad \vec{w} = \begin{pmatrix} 1+2i \\ 3i \\ -1 \end{pmatrix} \in \mathbb{C}^3.$$

You can no longer draw $\vec{u}$ as a single arrow in the plane: each entry is itself a point in the complex plane of Unit 1, so $\vec{u}$ has four real numbers inside it. The pictures in this unit are all of real vectors in $\mathbb{R}^2$, and they are there to build intuition. The algebra is what carries over to $\mathbb{C}^n$, and the algebra is identical except for one place (the inner product) where a complex conjugate appears.

Addition and scalar multiplication

Two vectors of the **same length** are added entry by entry:

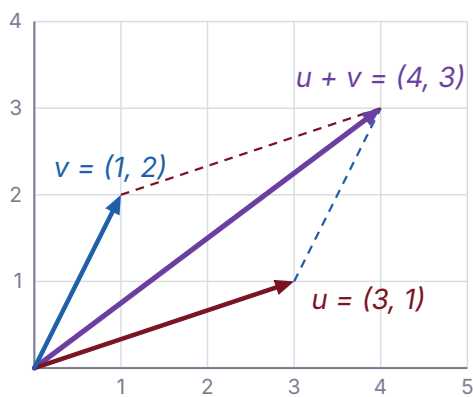
$$\vec{u} + \vec{v} = \begin{pmatrix} u_1 + v_1 \\ u_2 + v_2 \\ \vdots \\ u_n + v_n \end{pmatrix}.$$

A **scalar** is a single number, here any complex number. Multiplying a vector by a scalar c multiplies every entry by c :

$$c\vec{v} = \begin{pmatrix} c v_1 \\ c v_2 \\ \vdots \\ c v_n \end{pmatrix}.$$

The **zero vector** $\vec{0}$ has every entry zero. It is the only vector with $\vec{v} + \vec{0} = \vec{v}$ for every $\vec{v}$. Adding vectors of different lengths is undefined: there is no sensible answer to “ $(1, 2) + (1, 2, 3)$ ”.

ADDITION: TIP TO TAIL



SCALAR MULTIPLICATION

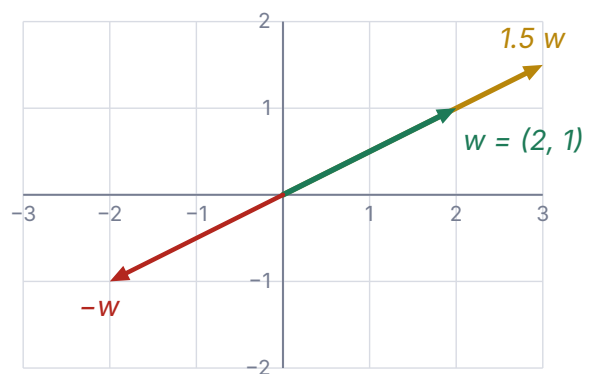


Figure 2.1 Left: adding $u = (3, 1)$ and $v = (1, 2)$ entry by entry gives $(4, 3)$, which is where you land by walking along u and then along v (or the other way round). Right: multiplying $w = (2, 1)$ by a real scalar keeps it on the same line through the origin. A factor of 1.5 stretches it, a factor of -1 flips it.

A complex scalar has no such simple picture. Multiplying by i rotates **each entry** by 90° in its own complex plane, which is not the same as rotating the arrow in Figure 2.1. Trust the entries, not the picture.

WORKED EXAMPLE · ARITHMETIC IN $\mathbb{C}^2$

Take $\vec{u} = (1 + i, 2)$ and $\vec{v} = (3, -i)$.

Sum. $\vec{u} + \vec{v} = (1 + i + 3, 2 - i) = (4 + i, 2 - i)$.

Complex scalar. $2i\vec{u} = (2i(1 + i), 2i \cdot 2) = (2i + 2i^2, 4i) = (-2 + 2i, 4i)$.

Combination. $2\vec{u} - i\vec{v} = (2 + 2i - 3i, 4 - i(-i)) = (2 - i, 4 + i^2) = (2 - i, 3)$.

The step to watch is $-i \cdot (-i) = i^2 = -1$. Sign slips in products of i are the most common error in this unit.

Linear combinations

A **linear combination** of vectors $\vec{v}_1, \dots, \vec{v}_k$ is any sum

$$c_1\vec{v}_1 + c_2\vec{v}_2 + \dots + c_k\vec{v}_k$$

with scalars $c_1, \dots, c_k$. The expression $2\vec{u} - i\vec{v}$ above is one. Linear combinations are the only thing you can build from vectors with addition and scaling, so almost every statement in this unit is about them. In Unit 3 the statement “a qubit is in a superposition of $|0\rangle$ and $|1\rangle$ ” will mean exactly “the state vector is a linear combination of two particular vectors”.

Transpose, conjugate and dagger

Three operations turn a vector into a related one.

DEFINITION · TRANSPOSE, CONJUGATE, CONJUGATE TRANSPOSE

The **transpose** $\vec{v}^T$ turns the column into a row with the same entries.

The **conjugate** $\vec{v}^*$ replaces every entry by its complex conjugate and keeps the column shape.

The **conjugate transpose**, or **dagger**, $\vec{v}^\dagger = (\vec{v}^*)^T$ does both: it turns the column into a row and conjugates every entry.

$$\vec{v} = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} \implies \vec{v}^\dagger = (v_1^* \quad v_2^*).$$

Some books write the dagger as A^H or A^* . This course uses $\dagger$ throughout, and $*$ only for the conjugate of a number or of each entry.

Applying the dagger twice gets you back where you started: $(\vec{v}^\dagger)^\dagger = \vec{v}$. For a real vector the conjugate changes nothing, so the dagger is just the transpose.

WORKED EXAMPLE · THE DAGGER OF A VECTOR

For $\vec{w} = (1 + 2i, 3i, -1)$, a column in $\mathbb{C}^3$:

$$\vec{w}^\dagger = (1 - 2i \quad -3i \quad -1).$$

Every entry is conjugated ($3i \rightarrow -3i$, and -1 is real so it stays) and the column has become a row. Forgetting either half is a mistake: $\vec{w}^T$ would keep $+3i$, and $\vec{w}^*$ would still be a column.

Why bother with rows at all? Because a row times a column is a single number (you will see the rule in the next section), and that number, $\vec{u}^\dagger \vec{v}$, is the inner product that the rest of the unit is built on.

NOTE · CHECK IT IN NUMPY

In Python the imaginary unit is written `1j`, so $1 + 2i$ is `1+2j`. A 1-D NumPy array does not remember whether it is a row or a column; that is fine for now.

```
import numpy as np
u = np.array([1+1j, 2])
v = np.array([3, -1j])
print(u + v)           # [4.+1.j 2.-1.j]
print(2*u - 1j*v)      # [2.-1.j 3.+0.j]
w = np.array([1+2j, 3j, -1])
print(w.conj())         # entries of the dagger: [1.-2.j 0.-3.j -1.-0.j]
```

Matrices

Session 5.

Shape and entries

A **matrix** is a rectangular grid of numbers. A matrix with m rows and n columns has **shape** $m \times n$ (rows first, always). The entry in row j and column k is written A_{jk} :

$$A = \begin{pmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \end{pmatrix} \text{ is } 2 \times 3.$$

A column vector in $\mathbb{C}^n$ is an $n \times 1$ matrix, and a row is a $1 \times n$ matrix. A **square** matrix has $m = n$. In quantum computing almost every matrix is square: 2×2 for one qubit, 4×4 for two.

Matrix times vector

If A is $m \times n$ and $\vec{v}$ has n entries, the product $A\vec{v}$ is a vector with m entries. There are two ways to compute it, and you should be fluent in both.

View 1: rows dotted with the vector. Entry j of $A\vec{v}$ is row j of A multiplied entry by entry with $\vec{v}$, then summed:

$$(A\vec{v})_j = \sum_{k=1}^n A_{jk} v_k.$$

View 2: a combination of the columns. Write the columns of A as $\vec{a}_1, \dots, \vec{a}_n$. Then

$$A\vec{v} = v_1\vec{a}_1 + v_2\vec{a}_2 + \dots + v_n\vec{a}_n.$$

The entries of $\vec{v}$ are the weights, and the columns of A are what gets weighted. View 1 is faster by hand. View 2 explains what a matrix *is*: its k -th column is where it sends the k -th standard basis vector $(0, \dots, 1, \dots, 0)$.

No conjugates appear anywhere in a matrix product. Complex entries are simply multiplied.

WORKED EXAMPLE · THE SAME PRODUCT, TWO WAYS

$$A = \begin{pmatrix} 1 & i \\ 2 & -1 \end{pmatrix}, \quad \vec{v} = \begin{pmatrix} i \\ 1 \end{pmatrix}.$$

Rows. First entry: $1 \cdot i + i \cdot 1 = 2i$. Second entry: $2 \cdot i + (-1) \cdot 1 = -1 + 2i$.

Columns.

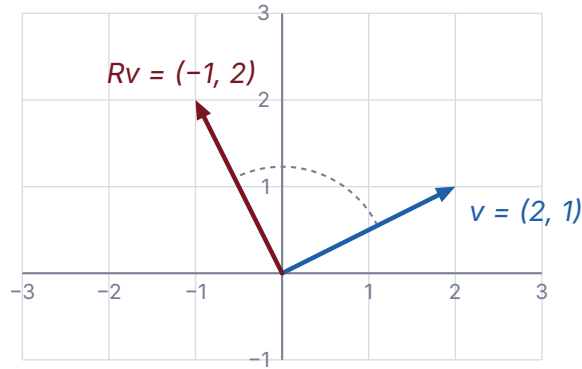
$$i \begin{pmatrix} 1 \\ 2 \end{pmatrix} + 1 \begin{pmatrix} i \\ -1 \end{pmatrix} = \begin{pmatrix} i+i \\ 2i-1 \end{pmatrix} = \begin{pmatrix} 2i \\ -1+2i \end{pmatrix}.$$

Both give $A\vec{v} = (2i, -1 + 2i)$. When two methods agree you can stop checking.

A matrix is an action

Because each output entry is a weighted sum of input entries, a matrix transforms vectors in a way that respects linear combinations: $A(c_1\vec{u} + c_2\vec{v}) = c_1A\vec{u} + c_2A\vec{v}$. Transformations with this property are called **linear**, which is where the name of the subject comes from. For real 2×2 matrices you can watch the action happen.

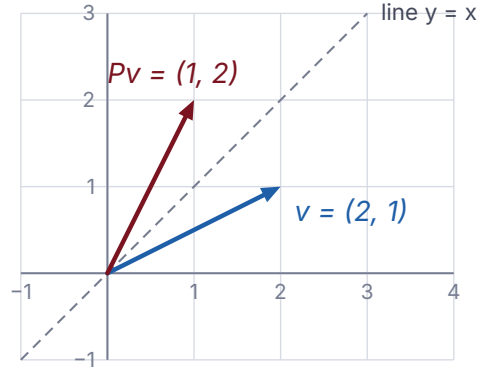
ROTATION BY 90°



$$R = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

Length unchanged: both vectors have norm $\sqrt{5}$.

SWAP OF ENTRIES



$$P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

A mirror image in the line $y = x$.

Figure 2.2 Two matrices acting on $v = (2, 1)$. Left: $R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ gives $Rv = (-1, 2)$, the same arrow turned anticlockwise by 90° . Right: $P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ swaps the two entries, $Pv = (1, 2)$, which is the mirror image in the line $y = x$.

Check the left panel with View 2. The first column of R is $(0, 1)$, so R sends $(1, 0)$ to $(0, 1)$, a quarter turn. The second column is $(-1, 0)$, so R sends $(0, 1)$ to $(-1, 0)$, also a quarter turn. A matrix that turns both basis arrows by 90° turns every arrow by 90° .

Matrix times matrix

If A is $m \times n$ and B is $n \times p$, the product AB is $m \times p$, with entries

$$(AB)_{jk} = \sum_{l=1}^n A_{jl} B_{lk}.$$

In words: entry (j, k) of AB is row j of A times column k of B . Equivalently, the k -th column of AB is A times the k -th column of B , so AB applied to a vector means "apply B first, then A ". **The matrix nearest the vector acts first.**

DEFINITION · WHEN IS A PRODUCT DEFINED?

AB exists only when the number of **columns of A** equals the number of **rows of B** . Write the shapes next to each other:

$$(m \times \underline{n}) (\underline{n} \times p) \longrightarrow m \times p.$$

The inner pair must match and the outer pair gives the shape of the answer. So a row times a column, $(1 \times n)(n \times 1)$, is 1×1 : a single number. A column times a row, $(n \times 1)(1 \times n)$, is an $n \times n$ matrix.

WORKED EXAMPLE · A COMPLEX MATRIX PRODUCT

$$A = \begin{pmatrix} 1 & i \\ 0 & 2 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 0 \\ 1 & -i \end{pmatrix}.$$

Row 1 of A with column 1 of B : $1 \cdot 2 + i \cdot 1 = 2 + i$. Row 1 with column 2: $1 \cdot 0 + i \cdot (-i) = -i^2 = 1$. Row 2 with column 1: $0 \cdot 2 + 2 \cdot 1 = 2$. Row 2 with column 2: $0 \cdot 0 + 2 \cdot (-i) = -2i$.

$$AB = \begin{pmatrix} 2+i & 1 \\ 2 & -2i \end{pmatrix}, \quad BA = \begin{pmatrix} 2 & 2i \\ 1 & -i \end{pmatrix}.$$

Work out BA yourself the same way. The two answers are different.

Order matters

For numbers, $ab = ba$ always. For matrices, AB and BA are usually different, and sometimes one of them is not even defined. This property is called **non-commutativity**, and in quantum mechanics it is a physical fact: doing operation B then A can leave a system in a different state from doing A then B .

WORKED EXAMPLE · TWO SIMPLE MATRICES THAT DO NOT COMMUTE

Take the swap P from Figure 2.2 and a matrix Q that flips the sign of the second entry:

$$P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Q = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$
$$PQ = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad QP = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

So $PQ \neq QP$; in fact $PQ = -QP$. Apply both to $(1, 0)$: Q leaves it alone and P swaps it to $(0, 1)$, so $PQ(1, 0) = (0, 1)$. The other order, QP , first swaps to $(0, 1)$ and then flips the sign, giving $(0, -1)$.

You will meet P and Q again in Unit 3 under their quantum-computing names.

PITFALL · READING A PRODUCT IN THE WRONG DIRECTION

$AB\vec{v}$ means $A(B\vec{v})$: B acts first. Students who read left to right as in a sentence apply A first and get the answer for $BA\vec{v}$. With commuting numbers that would not matter. With matrices it is simply a wrong answer.

The identity and the inverse

The **identity matrix** I has 1 on the diagonal (the entries $I_{11}, I_{22}, \dots$) and 0 everywhere else:

$$I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

It does nothing: $I\vec{v} = \vec{v}$ and $IA = AI = A$ for any matrix of matching size. The subscript gives the size when it matters.

A square matrix A has an **inverse** A^{-1} if $A^{-1}A = AA^{-1} = I$. The inverse undoes the action. For a 2×2 matrix there is a formula:

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \implies A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

Swap the diagonal entries, negate the off-diagonal ones, divide by $\det A = ad - bc$, the **determinant**. If $\det A = 0$ there is no inverse. That is all the determinant you need in this course (it returns once, in the eigenvalue section).

WORKED EXAMPLE · INVERTING TWO 2×2 MATRICES

Real. $A = \begin{pmatrix} 2 & 1 \\ 5 & 3 \end{pmatrix}$ has $\det A = 2 \cdot 3 - 1 \cdot 5 = 1$, so $A^{-1} = \begin{pmatrix} 3 & -1 \\ -5 & 2 \end{pmatrix}$. Check one entry of AA^{-1} : row 1 times column 1 is $2 \cdot 3 + 1 \cdot (-5) = 1$.

Complex. $C = \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix}$ has $\det C = 1 \cdot 1 - i \cdot i = 1 - i^2 = 2$, so

$$C^{-1} = \frac{1}{2} \begin{pmatrix} 1 & -i \\ -i & 1 \end{pmatrix}.$$

The same formula works for complex entries. The trap is $i \cdot i$: it is -1 , so the determinant is 2, not 0.

COMMON MISCONCEPTION · A MATRIX WITH NO ZERO ENTRIES ALWAYS HAS AN INVERSE

$\begin{pmatrix} 2 & 4 \\ 1 & 2 \end{pmatrix}$ has no zero entry and determinant $4 - 4 = 0$. Its second column is twice its first, so it squashes the whole plane onto one line, and a squashed plane cannot be un-squashed. It even sends a non-zero vector, $(2, -1)$, to $\vec{0}$.

Transpose and dagger of a matrix

The transpose A^T swaps rows and columns: $(A^T)_{jk} = A_{kj}$. An $m \times n$ matrix becomes $n \times m$. The dagger conjugates as well:

$$(A^\dagger)_{jk} = A_{kj}^*, \quad \begin{pmatrix} a & b \\ c & d \end{pmatrix}^\dagger = \begin{pmatrix} a^* & c^* \\ b^* & d^* \end{pmatrix}.$$

For a product, the order reverses:

$$(AB)^T = B^T A^T, \quad (AB)^\dagger = B^\dagger A^\dagger.$$

The shapes force the reversal. If A is 2×3 and B is 3×4 , then $A^\dagger$ is 3×2 and $B^\dagger$ is 4×3 , and only $B^\dagger A^\dagger$ is defined.

WORKED EXAMPLE · CHECKING $(AB)^\dagger = B^\dagger A^\dagger$

With A and B from the complex product above, $AB = \begin{pmatrix} 2+i & 1 \\ 2 & -2i \end{pmatrix}$, so

$$(AB)^\dagger = \begin{pmatrix} 2-i & 2 \\ 1 & 2i \end{pmatrix}.$$

Separately, $A^\dagger = \begin{pmatrix} 1 & 0 \\ -i & 2 \end{pmatrix}$ and $B^\dagger = \begin{pmatrix} 2 & 1 \\ 0 & i \end{pmatrix}$. Multiply in the reversed order: row 1 of $B^\dagger$ with column 1 of $A^\dagger$ is $2 \cdot 1 + 1 \cdot (-i) = 2 - i$, and the other three entries give 2, 1 and $2i$. The two matrices agree.

NOTE · CHECK IT IN NUMPY

Use `@` for matrix products. The `*` operator multiplies entry by entry, which is a different operation and a common source of silent errors.

```
A = np.array([[1, 1j], [0, 2]])
B = np.array([[2, 0], [1, -1j]])
print(A @ B)                # [[2+1j, 1], [2, -2j]]
print(np.allclose(A @ B, B @ A)) # False
dag = lambda M: M.conj().T
print(np.allclose(dag(A @ B), dag(B) @ dag(A))) # True
print(np.linalg.inv(np.array([[1, 1j], [1j, 1]])))
```

Inner product and norm

Session 6.

The definition

DEFINITION · INNER PRODUCT ON $\mathbb{C}^n$

For two vectors $\vec{u}, \vec{v} \in \mathbb{C}^n$,

$$\langle \vec{u}, \vec{v} \rangle = \vec{u}^\dagger \vec{v} = \sum_{k=1}^n u_k^* v_k = u_1^* v_1 + u_2^* v_2 + \cdots + u_n^* v_n.$$

The entries of the **first** vector are conjugated. The result is a single complex number.

The row-times-column rule from the previous section explains the notation: $\vec{u}^\dagger$ is a $1 \times n$ row, $\vec{v}$ is an $n \times 1$ column, and their product is 1×1 . For real vectors the conjugate does nothing and this is the dot product you know from school, $u_1 v_1 + u_2 v_2 + \cdots$.

WORKED EXAMPLE · AN INNER PRODUCT, BOTH WAYS ROUND

Take $\vec{u} = (1 + i, 2i)$ and $\vec{v} = (3, 1 - i)$.

$$\langle \vec{u}, \vec{v} \rangle = (1 + i)^* \cdot 3 + (2i)^*(1 - i) = (1 - i)3 + (-2i)(1 - i) = 3 - 3i - 2i + 2i^2 = 1 - 5i.$$

Now the other order: $\langle \vec{v}, \vec{u} \rangle = 3^*(1 + i) + (1 - i)^*(2i) = 3 + 3i + (1 + i)2i = 3 + 3i + 2i - 2 = 1 + 5i$.

The two answers are complex conjugates of each other. That is always true, and it is the first property below.

Why the conjugate is there

Every useful notion of length comes from the inner product of a vector with itself. So $\langle \vec{v}, \vec{v} \rangle$ had better be a real number that is never negative, and zero only for $\vec{v} = \vec{0}$.

The conjugate is what guarantees this. With it,

$$\langle \vec{v}, \vec{v} \rangle = \sum_k v_k^* v_k = \sum_k |v_k|^2,$$

because $z^* z = |z|^2$ from Unit 1. A sum of squared moduli is real and at least zero.

WORKED EXAMPLE · WHAT GOES WRONG WITHOUT THE CONJUGATE

Take $\vec{w} = (1, i)$. Clearly not the zero vector.

Without conjugating: $1 \cdot 1 + i \cdot i = 1 + i^2 = 0$. The vector would have length zero, which is absurd.

With the conjugate: $1^* \cdot 1 + i^* \cdot i = 1 + (-i)(i) = 1 + 1 = 2$. So $\vec{w}$ has length $\sqrt{2}$, which is what the moduli say: $|1|^2 + |i|^2 = 2$.

Try $\vec{w} = (1, 2i)$ without the conjugate and you get $1 - 4 = -3$, a negative squared length. The school dot product simply does not work for complex vectors.

Properties

For all $\vec{u}, \vec{v}, \vec{w} \in \mathbb{C}^n$ and every complex scalar c :

1. **Conjugate symmetry.** $\langle \vec{v}, \vec{u} \rangle = \langle \vec{u}, \vec{v} \rangle^*$.
2. **Linear in the second argument.** $\langle \vec{u}, c\vec{v} + \vec{w} \rangle = c\langle \vec{u}, \vec{v} \rangle + \langle \vec{u}, \vec{w} \rangle$.
3. **Positive.** $\langle \vec{v}, \vec{v} \rangle$ is real and ≥ 0 , with equality only when $\vec{v} = \vec{0}$.

Put 1 and 2 together and you find what happens to a scalar in the **first** slot: it comes out conjugated,

$$\langle c\vec{u}, \vec{v} \rangle = c^* \langle \vec{u}, \vec{v} \rangle.$$

Each property follows in a line from the sum $\sum_k u_k^* v_k$. For instance, the conjugate of $\sum_k u_k^* v_k$ is $\sum_k u_k v_k^* = \sum_k v_k^* u_k = \langle \vec{v}, \vec{u} \rangle$, which is property 1.

PITFALL · MATHS BOOKS CONJUGATE THE OTHER ARGUMENT

Many mathematics textbooks, and some linear-algebra courses you may take later, define $\langle \vec{u}, \vec{v} \rangle = \sum_k u_k v_k^*$, conjugating the **second** vector. Their inner product is the complex conjugate of ours, and it is linear in the first slot instead of the second.

Norms and orthogonality come out the same either way, so the difference hides until you compute a coefficient: then the maths convention gives you the conjugate of the physics answer. Everything in this course, and every quantum textbook, uses the physics convention: **conjugate the first argument**. When you read another source, check which one it uses before copying a formula.

Norm and normalisation

DEFINITION · NORM AND UNIT VECTOR

The **norm** (length) of $\vec{v}$ is

$$\|\vec{v}\| = \sqrt{\langle \vec{v}, \vec{v} \rangle} = \sqrt{|v_1|^2 + |v_2|^2 + \cdots + |v_n|^2}.$$

A **unit vector** has norm 1. **Normalising** a non-zero vector means dividing it by its norm, $\hat{v} = \vec{v}/\|\vec{v}\|$, which gives a unit vector pointing the same way.

Scaling a vector scales its norm by the modulus of the scalar: $\|c\vec{v}\| = |c| \|\vec{v}\|$. In particular, multiplying a unit vector by a phase $e^{i\theta}$ from Unit 1 gives another unit vector, since $|e^{i\theta}| = 1$.

WORKED EXAMPLE · NORMALISING A COMPLEX VECTOR

Take $\vec{v} = (3, 4i)$. The squared moduli are $|3|^2 = 9$ and $|4i|^2 = 16$, so $\|\vec{v}\| = \sqrt{25} = 5$ and

$$\hat{v} = \frac{1}{5} \begin{pmatrix} 3 \\ 4i \end{pmatrix} = \begin{pmatrix} 3/5 \\ 4i/5 \end{pmatrix}.$$

Check: $(3/5)^2 + (4/5)^2 = 9/25 + 16/25 = 1$.

The mistake to avoid is squaring $4i$ to get -16 . The norm uses $|4i|^2$, which is 16.

In Unit 3 every quantum state is a unit vector. When a problem hands you a state that is not normalised, the first step is always to normalise it.

Orthogonality

DEFINITION · ORTHOGONAL VECTORS

$\vec{u}$ and $\vec{v}$ are **orthogonal** if $\langle \vec{u}, \vec{v} \rangle = 0$.

For real vectors in the plane this means perpendicular. For complex vectors there is no picture, and the definition is the inner product. Because $\langle \vec{v}, \vec{u} \rangle = \langle \vec{u}, \vec{v} \rangle^*$, and the conjugate of 0 is 0, the order does not matter for this test.

WORKED EXAMPLE · TWO COMPLEX VECTORS AT RIGHT ANGLES

Take $\vec{u} = (1, i)$ and $\vec{v} = (1, -i)$.

$$\langle \vec{u}, \vec{v} \rangle = 1^* \cdot 1 + i^* \cdot (-i) = 1 + (-i)(-i) = 1 + i^2 = 0.$$

They are orthogonal. Had you forgotten the conjugate you would have computed $1 + i(-i) = 2$ and concluded they were not. Each has norm $\sqrt{2}$, so $\frac{1}{\sqrt{2}}(1, i)$ and $\frac{1}{\sqrt{2}}(1, -i)$ are two orthogonal unit vectors. You will see this pair again in Unit 3.

NOTE · CHECK IT IN NUMPY

`np.vdot(u, v)` conjugates its **first** argument, exactly like our inner product. `np.dot(u, v)` and `u @ v` do **not** conjugate, so they compute the broken version from the worked example above.

```
u = np.array([1+1j, 2j]); v = np.array([3, 1-1j])
print(np.vdot(u, v))           # (1-5j)
w = np.array([1, 1j])
print(w @ w, np.vdot(w, w))    # 0j (wrong) (2+0j) (right)
print(np.linalg.norm([3, 4j])) # 5.0
print(np.vdot([1, 1j], [1, -1j])) # 0j: orthogonal
```

Bases and projection

Session 7.

Linear independence and span

A set of vectors is **linearly independent** if none of them can be built as a linear combination of the others. Informally, each one contributes a direction the rest do not already cover. In $\mathbb{R}^2$, the vectors $(1, 0)$ and $(2, 0)$ are not independent (the second is twice the first), while $(1, 0)$ and $(1, 1)$ are.

The **span** of a set of vectors is the collection of all their linear combinations. $(1, 0)$ and $(1, 1)$ span the whole plane: any (a, b) equals $(a - b)(1, 0) + b(1, 1)$.

DEFINITION · BASIS

A **basis** of $\mathbb{C}^n$ is a set of linearly independent vectors that spans $\mathbb{C}^n$. Every basis of $\mathbb{C}^n$ has exactly n vectors, and every vector can be written as a linear combination of basis vectors in exactly one way. The weights in that combination are the vector's **components** in that basis.

The standard basis

The simplest basis of $\mathbb{C}^n$ is the **standard basis**: $\vec{e}_k$ has a 1 in entry k and zeros elsewhere. In $\mathbb{C}^2$,

$$\vec{e}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \vec{e}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = v_1 \vec{e}_1 + v_2 \vec{e}_2.$$

Components in the standard basis are simply the entries. That is why a column of numbers feels like "the" vector. It is really the vector's description in one particular basis, and much of quantum mechanics is about changing to another.

Orthonormal bases

DEFINITION · ORTHONORMAL BASIS

A basis $\{\vec{e}_1, \dots, \vec{e}_n\}$ is **orthonormal** if every vector has norm 1 and every pair is orthogonal:

$$\langle \vec{e}_j, \vec{e}_k \rangle = \begin{cases} 1 & j = k \\ 0 & j \neq k \end{cases}$$

The standard basis is orthonormal. So are many others. Two will run through the rest of this unit and into Unit 3.

The diagonal basis of $\mathbb{R}^2$.

$$\vec{f}_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \vec{f}_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

Each has norm $\sqrt{\frac{1}{2} + \frac{1}{2}} = 1$, and $\langle \vec{f}_1, \vec{f}_2 \rangle = \frac{1}{2}(1 - 1) = 0$. These are the standard basis turned by 45° (and one of them flipped).

A complex orthonormal basis of $\mathbb{C}^2$.

$$\vec{g}_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}, \quad \vec{g}_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix}.$$

You checked in the previous section that these are orthogonal unit vectors. This basis has no real picture, and it is every bit as good a basis as the other two.

PITFALL · ORTHOGONAL IS NOT ORTHONORMAL

$(1, 1)$ and $(1, -1)$ are orthogonal, and they form a basis, but each has norm $\sqrt{2}$. The expansion formula below needs **unit** vectors. Using unnormalised vectors in it gives components that are too large by exactly the missing factor.

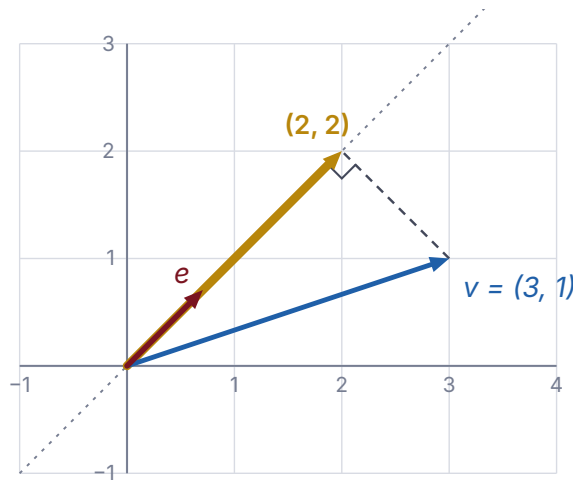
The inner product as projection

Here is the geometric meaning of the inner product, and the main idea of this unit.

DEFINITION · COMPONENT ALONG A UNIT VECTOR

If $\vec{e}$ is a **unit** vector, the **component** of $\vec{v}$ along $\vec{e}$ is the number $\langle \vec{e}, \vec{v} \rangle$, and the **projection** of $\vec{v}$ onto $\vec{e}$ is the vector $\langle \vec{e}, \vec{v} \rangle \vec{e}$.

In the real plane you can see it. Drop a perpendicular from the tip of $\vec{v}$ onto the line through $\vec{e}$. The projection is the arrow from the origin to the foot of that perpendicular, and the component is its signed length.



UNIT VECTOR

$$e = (1, 1) / \sqrt{2}, \|e\| = 1$$

COMPONENT OF v ALONG e

$$\langle e, v \rangle = (3 + 1) / \sqrt{2} = 2\sqrt{2} \approx 2.83$$

the length of the gold arrow

PROJECTION VECTOR

$$\langle e, v \rangle e = 2\sqrt{2} \cdot (1, 1) / \sqrt{2} = (2, 2)$$

WHAT IS LEFT OVER

$$v - (2, 2) = (1, -1), \text{ at right angles to } e$$

Figure 2.3 Projecting $v = (3, 1)$ onto the unit vector $e = \frac{1}{\sqrt{2}}(1, 1)$. The component $\langle e, v \rangle = 2\sqrt{2}$ is the length of the gold arrow, and the projection $2\sqrt{2}e = (2, 2)$ is the gold arrow itself. What is left, $v - (2, 2) = (1, -1)$, meets the line at a right angle.

Why the leftover is perpendicular, in one line: with $\vec{p} = \langle \vec{e}, \vec{v} \rangle \vec{e}$,

$$\langle \vec{e}, \vec{v} - \vec{p} \rangle = \langle \vec{e}, \vec{v} \rangle - \langle \vec{e}, \vec{v} \rangle \langle \vec{e}, \vec{e} \rangle = 0,$$

using linearity in the second slot and $\langle \vec{e}, \vec{e} \rangle = 1$. That is the whole reason $\vec{e}$ must be a unit vector.

WORKED EXAMPLE · A PROJECTION IN THE PLANE

Component of $\vec{v} = (3, 1)$ along $\vec{e} = \frac{1}{\sqrt{2}}(1, 1)$:

$$\langle \vec{e}, \vec{v} \rangle = \frac{1}{\sqrt{2}}(1 \cdot 3 + 1 \cdot 1) = \frac{4}{\sqrt{2}} = 2\sqrt{2} \approx 2.83.$$

Projection: $2\sqrt{2} \cdot \frac{1}{\sqrt{2}}(1, 1) = (2, 2)$. Leftover: $(3, 1) - (2, 2) = (1, -1)$, and $\langle \vec{e}, (1, -1) \rangle = \frac{1}{\sqrt{2}}(1 - 1) = 0$. Everything matches Figure 2.3.

Expanding a vector in any orthonormal basis

Project onto every vector of an orthonormal basis and add the projections back up. You recover the vector.

DEFINITION · EXPANSION IN AN ORTHONORMAL BASIS

If $\{\vec{e}_1, \dots, \vec{e}_n\}$ is an orthonormal basis of $\mathbb{C}^n$, then for every $\vec{v}$,

$$\vec{v} = \sum_{k=1}^n \langle \vec{e}_k, \vec{v} \rangle \vec{e}_k.$$

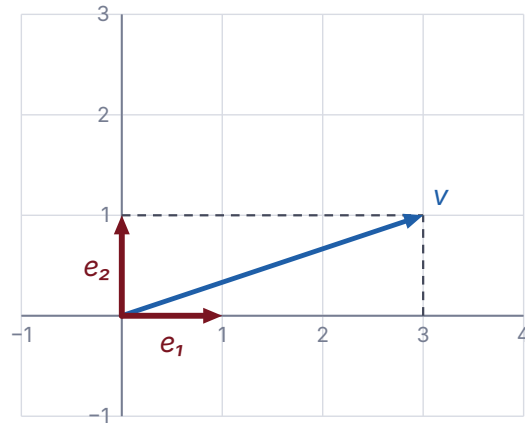
The components are $c_k = \langle \vec{e}_k, \vec{v} \rangle$: basis vector first, the vector you are expanding second.

The proof is short enough to remember. Any $\vec{v}$ can be written as $\vec{v} = \sum_k c_k \vec{e}_k$ for some numbers c_k , because the $\vec{e}_k$ form a basis. Take the inner product with $\vec{e}_j$ and use linearity in the second slot:

$$\langle \vec{e}_j, \vec{v} \rangle = \sum_k c_k \langle \vec{e}_j, \vec{e}_k \rangle = c_j,$$

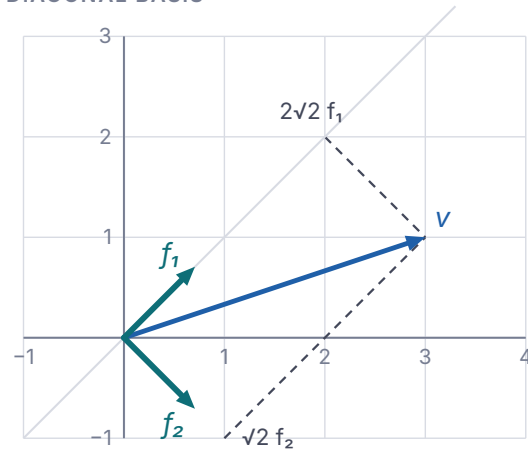
since only the term with $k = j$ survives. Notice where the physics convention paid off: with the basis vector in the first slot, c_j comes out without a conjugate.

STANDARD BASIS



$$v = 3 e_1 + 1 e_2 \quad \cdot \quad 3^2 + 1^2 = 10$$

DIAGONAL BASIS



$$v = 2\sqrt{2} f_1 + \sqrt{2} f_2 \quad \cdot \quad 8 + 2 = 10$$

Figure 2.4 One vector, two descriptions. In the standard basis, $v = (3, 1)$ has components 3 and 1. In the diagonal basis $f_1 = \frac{1}{\sqrt{2}}(1, 1)$, $f_2 = \frac{1}{\sqrt{2}}(1, -1)$ the same arrow has components $2\sqrt{2}$ and $\sqrt{2}$. The arrow did not move; only the grid used to describe it changed. In both cases the squared components add to $\|v\|^2 = 10$.

WORKED EXAMPLE · CHANGING TO THE DIAGONAL BASIS

Expand $\vec{v} = (3, 1)$ in $\{\vec{f}_1, \vec{f}_2\}$.

$$c_1 = \langle \vec{f}_1, \vec{v} \rangle = \frac{1}{\sqrt{2}}(3 + 1) = 2\sqrt{2}, \quad c_2 = \langle \vec{f}_2, \vec{v} \rangle = \frac{1}{\sqrt{2}}(3 - 1) = \sqrt{2}.$$

Rebuild: $2\sqrt{2} \cdot \frac{1}{\sqrt{2}}(1, 1) + \sqrt{2} \cdot \frac{1}{\sqrt{2}}(1, -1) = (2, 2) + (1, -1) = (3, 1)$.

WORKED EXAMPLE · CHANGING TO A COMPLEX BASIS

Expand $\vec{v} = (2, 1 + i)$ in $\{\vec{g}_1, \vec{g}_2\}$. Now the conjugate on the first argument matters, because the basis vectors are complex.

$$c_1 = \langle \vec{g}_1, \vec{v} \rangle = \frac{1}{\sqrt{2}}(1 \cdot 2 + (-i)(1 + i)) = \frac{1}{\sqrt{2}}(2 - i + 1) = \frac{1}{\sqrt{2}}(3 - i),$$

$$c_2 = \langle \vec{g}_2, \vec{v} \rangle = \frac{1}{\sqrt{2}}(1 \cdot 2 + (i)(1 + i)) = \frac{1}{\sqrt{2}}(2 + i - 1) = \frac{1}{\sqrt{2}}(1 + i).$$

Rebuild the first entry: $\frac{1}{2}((3 - i) + (1 + i)) = 2$. The second: $\frac{1}{2}((3 - i)i + (1 + i)(-i)) = \frac{1}{2}(3i + 1 - i + 1) = 1 + i$. Correct.

Pythagoras in $\mathbb{C}^n$

In Figure 2.4 the squared components add up to the same number in both bases: $3^2 + 1^2 = 10$ and $(2\sqrt{2})^2 + (\sqrt{2})^2 = 8 + 2 = 10$. That is Pythagoras, and it holds in every orthonormal basis of $\mathbb{C}^n$:

$$\|\vec{v}\|^2 = \sum_{k=1}^n |\langle \vec{e}_k, \vec{v} \rangle|^2.$$

To see why, substitute $\vec{v} = \sum_k c_k \vec{e}_k$ into $\langle \vec{v}, \vec{v} \rangle$. Every cross term contains $\langle \vec{e}_j, \vec{e}_k \rangle = 0$ for $j \neq k$, and what survives is $\sum_k c_k^* c_k = \sum_k |c_k|^2$.

Check it on the complex example: $|c_1|^2 = \frac{1}{2}(9 + 1) = 5$ and $|c_2|^2 = \frac{1}{2}(1 + 1) = 1$, total 6, while $\|\vec{v}\|^2 = |2|^2 + |1 + i|^2 = 4 + 2 = 6$.

For a unit vector the numbers $|\langle \vec{e}_k, \vec{v} \rangle|^2$ are non-negative and add up to 1, which is exactly what a list of probabilities must do: in Unit 3 they become the probabilities of the measurement outcomes, and the rule that says so is called the Born rule.

NOTE · CHECK IT IN NUMPY

```
g1 = np.array([1, 1j]) / np.sqrt(2)
g2 = np.array([1, -1j]) / np.sqrt(2)
v = np.array([2, 1+1j])
c1, c2 = np.vdot(g1, v), np.vdot(g2, v)
print(c1 * np.sqrt(2), c2 * np.sqrt(2))      # (3-1j) (1+1j)
print(np.allclose(c1*g1 + c2*g2, v))         # True
print(abs(c1)**2 + abs(c2)**2, np.vdot(v, v).real)  # 6.0 6.0
```

Special matrices

Session 8.

A few families of square matrices carry most of quantum mechanics. Each is defined by a one-line condition, and each condition buys you a guarantee.

Diagonal and symmetric matrices

A **diagonal** matrix has zeros everywhere off the main diagonal, such as $\begin{pmatrix} 2 & 0 \\ 0 & -1 \end{pmatrix}$. It scales each standard basis vector by its own factor and does nothing else. Diagonal matrices are the easiest to work with: products, powers and inverses are all done entry by entry.

A **symmetric** matrix equals its own transpose, $A^T = A$, so $A_{jk} = A_{kj}$. It is a mirror image of itself across the diagonal, such as $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$. For complex matrices, symmetry is the wrong condition, and its replacement is the next definition.

Hermitian matrices

DEFINITION · HERMITIAN MATRIX

A square matrix M is **Hermitian** if $M^\dagger = M$, that is, $M_{jk} = M_{kj}^*$ for every j, k .

Two consequences follow straight from $M_{jk} = M_{kj}^*$:

- Every **diagonal entry is real**, because $M_{kk} = M_{kk}^*$ and only real numbers equal their own conjugate.
- The entries **across the diagonal are conjugates** of each other.

A real symmetric matrix is automatically Hermitian, because conjugating a real number changes nothing.

WORKED EXAMPLE · RECOGNISING A HERMITIAN MATRIX

$$M = \begin{pmatrix} 2 & 1-i \\ 1+i & 3 \end{pmatrix}.$$

Conjugate every entry: $\begin{pmatrix} 2 & 1+i \\ 1-i & 3 \end{pmatrix}$. Transpose: $\begin{pmatrix} 2 & 1-i \\ 1+i & 3 \end{pmatrix}$. That is M again, so $M^\dagger = M$. The diagonal (2 and 3) is real, and the off-diagonal pair $1-i, 1+i$ are conjugates.

By contrast $\begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix}$ is symmetric but **not** Hermitian: its dagger has $-i$ where it has i .

A Hermitian matrix can be moved from one side of an inner product to the other: $\langle \vec{u}, M\vec{v} \rangle = \langle M\vec{u}, \vec{v} \rangle$. (Proof: $(M\vec{u})^\dagger \vec{v} = \vec{u}^\dagger M^\dagger \vec{v} = \vec{u}^\dagger M\vec{v}$.) This small fact is behind every nice property Hermitian matrices have, and Hermitian matrices are what quantum mechanics uses for measurable quantities.

Unitary matrices

DEFINITION · UNITARY MATRIX

A square matrix U is **unitary** if $U^\dagger U = I$. Equivalently, U has an inverse and $U^{-1} = U^\dagger$.

The condition looks abstract. What it guarantees is concrete: **a unitary matrix preserves inner products**, and therefore norms and orthogonality.

$$\langle U\vec{u}, U\vec{v} \rangle = (U\vec{u})^\dagger (U\vec{v}) = \vec{u}^\dagger U^\dagger U \vec{v} = \vec{u}^\dagger \vec{v} = \langle \vec{u}, \vec{v} \rangle.$$

Set $\vec{u} = \vec{v}$ and you get $\|U\vec{v}\| = \|\vec{v}\|$: a unitary matrix sends unit vectors to unit vectors. That is why, in Unit 3, every quantum operation is unitary. A quantum state must stay a unit vector, and unitaries are the matrices that guarantee it.

There is also a way to spot a unitary by eye. Entry (j, k) of $U^\dagger U$ is $\langle \vec{u}_j, \vec{u}_k \rangle$, where $\vec{u}_j$ is column j of U . So $U^\dagger U = I$ says exactly this: **the columns of a unitary matrix form an orthonormal basis**.

WORKED EXAMPLE · THREE UNITARY MATRICES

A rotation. $R(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$. Its columns are unit vectors (because $\cos^2 \theta + \sin^2 \theta = 1$) and orthogonal (because $-\cos \theta \sin \theta + \sin \theta \cos \theta = 0$). So $R(\theta)$ is unitary. The matrix R of Figure 2.2 is $R(90^\circ)$. A real unitary matrix is also called **orthogonal**.

A phase. $\begin{pmatrix} 1 & 0 \\ 0 & e^{i\varphi} \end{pmatrix}$. Its dagger is $\begin{pmatrix} 1 & 0 \\ 0 & e^{-i\varphi} \end{pmatrix}$, and the product is I because $e^{-i\varphi} e^{i\varphi} = 1$. It leaves the first entry of a vector alone and rotates the phase of the second.

A complex mixer. $V = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix}$. Its columns $\frac{1}{\sqrt{2}}(1, i)$ and $\frac{1}{\sqrt{2}}(i, 1)$ have norm 1, and $\langle (1, i), (i, 1) \rangle = 1 \cdot i + (-i) \cdot 1 = 0$. So V is unitary. It sends $(1, 0)$ to $\frac{1}{\sqrt{2}}(1, i)$, still a unit vector.

COMMON MISCONCEPTION · ANY INVERTIBLE MATRIX PRESERVES LENGTH

$N = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ has determinant 1, so it is invertible. But $N(0, 1) = (1, 1)$: a vector of norm 1 becomes a vector of norm $\sqrt{2}$. Invertible means the action can be undone. Unitary is the much stronger statement that no length or angle changes at all.

Summary of the special matrices

TYPE	CONDITION	DIAGONAL ENTRIES	EIGENVALUES	PRESERVES NORMS?	EXAMPLE
Identity	$I_{jk} = 1$ if $j = k$, else 0	all 1	all 1	yes	I_2
Diagonal	zero off the diagonal	anything	the diagonal entries	only if each $ d_k = 1$	$\begin{pmatrix} 2 & 0 \\ 0 & -1 \end{pmatrix}$
Symmetric	$A^T = A$	anything	real if A is real	not in general	$\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$
Hermitian	$M^\dagger = M$	real	real	not in general	$\begin{pmatrix} 2 & 1-i \\ 1+i & 3 \end{pmatrix}$
Unitary	$U^\dagger U = I$	modulus at most 1	modulus 1	yes	$\begin{pmatrix} 1 & 0 \\ 0 & e^{i\varphi} \end{pmatrix}$

The eigenvalue column is explained in the next section. Some matrices belong to several rows at once: I is in all five, and $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ is symmetric, Hermitian and unitary.

NOTE · CHECK IT IN NUMPY

```
dag = lambda M: M.conj().T
M = np.array([[2, 1-1j], [1+1j, 3]])
V = np.array([[1, 1j], [1j, 1]]) / np.sqrt(2)
print(np.allclose(dag(M), M))           # True: Hermitian
print(np.allclose(dag(V) @ V, np.eye(2))) # True: unitary
x = np.array([0.6, 0.8j])
print(np.linalg.norm(V @ x))           # 1.0, norm kept
```

Eigenvalues and eigenvectors

Session 9.

BEYOND THE SYLLABUS · WHY THIS IS HERE

Eigenvalues are not in the printed syllabus. They are added because Unit 3 cannot be done without them: the possible results of a quantum measurement are the eigenvalues of a Hermitian matrix, and the states that give those results with certainty are its eigenvectors. This section is taught and examinable.

The definition

Most vectors change direction when a matrix acts on them. A few special ones do not: the matrix only scales them.

DEFINITION · EIGENVALUE AND EIGENVECTOR

A non-zero vector $\vec{v}$ is an **eigenvector** of the square matrix A , with **eigenvalue** λ (a complex number), if

$$A\vec{v} = \lambda\vec{v}.$$

The zero vector is excluded, since $A\vec{0} = \lambda\vec{0}$ for every λ and that says nothing. Any non-zero multiple of an eigenvector is again an eigenvector with the same eigenvalue, because $A(c\vec{v}) = cA\vec{v} = \lambda(c\vec{v})$. So eigenvectors are only defined up to a scalar factor, and it is usual to normalise them.

You have already seen examples. In Figure 2.2, the swap P leaves $(1, 1)$ exactly where it is ($\lambda = 1$) and sends $(1, -1)$ to $(-1, 1)$, which is -1 times itself ($\lambda = -1$). Both vectors lie along the mirror line or perpendicular to it.

Finding them for a 2×2 matrix

Rewrite $A\vec{v} = \lambda\vec{v}$ as $(A - \lambda I)\vec{v} = \vec{0}$. A non-zero $\vec{v}$ can be sent to $\vec{0}$ only if $A - \lambda I$ has no inverse, that is, if its determinant is zero. For

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad \det(A - \lambda I) = (a - \lambda)(d - \lambda) - bc = 0,$$

which expands to the **characteristic equation**

$$\lambda^2 - (a + d)\lambda + (ad - bc) = 0.$$

It is a quadratic, so there are two roots (possibly equal, possibly complex). Solve it with the quadratic formula, then for each root λ solve $(A - \lambda I)\vec{v} = \vec{0}$. The two equations in that system are always multiples of each other, so use either one.

WORKED EXAMPLE · A REAL SYMMETRIC MATRIX

Take $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$. Here $a + d = 4$ and $ad - bc = 3$:

$$\lambda^2 - 4\lambda + 3 = (\lambda - 1)(\lambda - 3) = 0 \implies \lambda = 1 \text{ or } 3.$$

$\lambda = 3$. The first row of $A - 3I$ gives $-x + y = 0$, so $y = x$. Eigenvector $(1, 1)$. Check: $A(1, 1) = (3, 3)$.

$\lambda = 1$. The first row of $A - I$ gives $x + y = 0$. Eigenvector $(1, -1)$. Check: $A(1, -1) = (1, -1)$.

Normalised, these are $\vec{f}_1$ and $\vec{f}_2$, the diagonal basis of the previous section. The eigenvalues are real and the eigenvectors are orthogonal.

WORKED EXAMPLE · A HERMITIAN MATRIX WITH COMPLEX ENTRIES

Take $M = \begin{pmatrix} 2 & 1 - i \\ 1 + i & 3 \end{pmatrix}$. Here $a + d = 5$ and $ad - bc = 6 - (1 - i)(1 + i) = 6 - 2 = 4$:

$$\lambda^2 - 5\lambda + 4 = (\lambda - 1)(\lambda - 4) = 0 \implies \lambda = 1 \text{ or } 4.$$

$\lambda = 4$. First row: $-2x + (1 - i)y = 0$. Take $y = 2$, then $x = 1 - i$. Eigenvector $(1 - i, 2)$. Check the second entry of $M\vec{v}$: $(1 + i)(1 - i) + 3 \cdot 2 = 2 + 6 = 8 = 4 \cdot 2$.

$\lambda = 1$. First row: $x + (1 - i)y = 0$. Take $y = 1$, then $x = -1 + i$. Eigenvector $(-1 + i, 1)$.

Orthogonal? $\langle (1 - i, 2), (-1 + i, 1) \rangle = (1 + i)(-1 + i) + 2 \cdot 1 = -2 + 2 = 0$. Yes. The norms are $\sqrt{6}$ and $\sqrt{3}$, so the normalised eigenvectors are $\frac{1}{\sqrt{6}}(1 - i, 2)$ and $\frac{1}{\sqrt{3}}(-1 + i, 1)$.

Two facts to remember

DEFINITION · EIGENVALUES OF HERMITIAN AND UNITARY MATRICES

Hermitian. The eigenvalues of a Hermitian matrix are **real**, and eigenvectors with different eigenvalues are **orthogonal**. After normalising, the eigenvectors of an $n \times n$ Hermitian matrix can always be chosen to form an orthonormal basis.

Unitary. Every eigenvalue of a unitary matrix has **modulus 1**, so it can be written $e^{i\theta}$.

The unitary fact takes one line. If $U\vec{v} = \lambda\vec{v}$, then $\|\vec{v}\| = \|U\vec{v}\| = \|\lambda\vec{v}\| = |\lambda| \|\vec{v}\|$, and dividing by $\|\vec{v}\| \neq 0$ gives $|\lambda| = 1$. The Hermitian fact about real eigenvalues is almost as short: $\lambda \|\vec{v}\|^2 = \langle \vec{v}, M\vec{v} \rangle = \langle M\vec{v}, \vec{v} \rangle = \lambda^* \|\vec{v}\|^2$, so $\lambda = \lambda^*$.

WORKED EXAMPLE · A REAL MATRIX WITH COMPLEX EIGENVALUES

Take $R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, the quarter turn. Here $a + d = 0$ and $ad - bc = 1$, so $\lambda^2 + 1 = 0$ and $\lambda = \pm i$. Both have modulus 1, as they must for a unitary matrix.

For $\lambda = i$: $-ix - y = 0$, so $y = -ix$ and an eigenvector is $(1, -i)$. Check: $R(1, -i) = (i, 1) = i(1, -i)$.

A quarter turn moves every real arrow, so it has no real eigenvector. Its eigenvectors are complex. This is one more place where the real numbers are not enough.

PITFALL · NUMPY'S EIGENVECTORS LOOK DIFFERENT FROM YOURS

`np.linalg.eigh(M)` returns the eigenvalues in increasing order and the **normalised eigenvectors as the columns** of a matrix. Each column may differ from your hand answer by a factor such as -1 or $e^{i\theta}$, since any multiple of an eigenvector is still an eigenvector. Compare by checking $M\vec{v} = \lambda\vec{v}$, never by comparing entries. Use `eigh` only for Hermitian matrices; for others use `np.linalg.eig`.

```
M = np.array([[2, 1-1j], [1+1j, 3]])
vals, vecs = np.linalg.eigh(M)
print(vals)                # [1. 4.]
v = vecs[:, 1]              # column for eigenvalue 4
print(np.allclose(M @ v, 4 * v))  # True
print(np.linalg.eig(np.array([[0, -1], [1, 0]]))[0])  # [0.+1.j 0.-1.j]
```

Why Unit 3 needs this

In Unit 3, a measurable quantity (an **observable**) is a Hermitian matrix. The possible measured values are its eigenvalues, which is why it matters that they are real: a meter never reads $2 + 3i$. The eigenvectors form an orthonormal basis, and measuring the quantity is measuring in that basis, with probabilities given by the projections of the previous section. Unitary matrices, whose eigenvalues all have modulus 1, will be the gates.

The tensor product

Session 9, continued.

BEYOND THE SYLLABUS · WHY THIS IS HERE

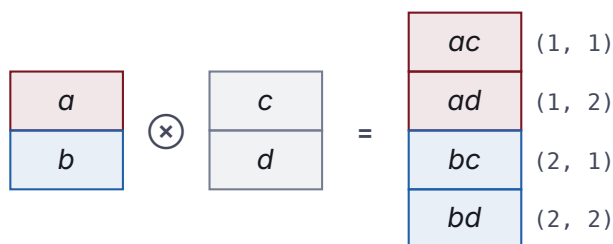
The tensor product is not in the printed syllabus either. It is added because the syllabus item “multiple quantum states” in Unit 3 is impossible without it: the state of two qubits lives in the tensor product of two copies of $\mathbb{C}^2$. Taught and examinable.

Tensor product of two vectors

For $\vec{u} \in \mathbb{C}^2$ and $\vec{v} \in \mathbb{C}^2$, the **tensor product** (in components, the **Kronecker product**) is the vector in $\mathbb{C}^4$ formed by multiplying every entry of $\vec{u}$ by the whole of $\vec{v}$ and stacking the results:

$$\begin{pmatrix} a \\ b \end{pmatrix} \otimes \begin{pmatrix} c \\ d \end{pmatrix} = \begin{pmatrix} a \begin{pmatrix} c \\ d \end{pmatrix} \\ b \begin{pmatrix} c \\ d \end{pmatrix} \end{pmatrix} = \begin{pmatrix} ac \\ ad \\ bc \\ bd \end{pmatrix}.$$

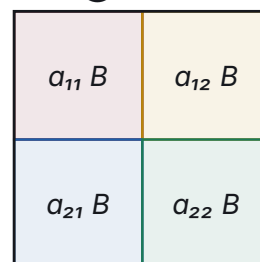
TWO VECTORS



Top half: a times (c, d) . Bottom half: b times (c, d) .
Entry label (j, k) : j -th entry of the first factor,

TWO MATRICES

$A \otimes B$ (4×4)



Each block is a 2×2 copy of B , scaled by one entry of A .

$(2 \times 2) \otimes (2 \times 2)$ gives 4×4 .

$(m \times n) \otimes (p \times q)$ gives $mp \times nq$.

Figure 2.5 Left: the four entries of $u \otimes v$, labelled by which entry of each factor they came from. The first factor's index changes slowest. Right: the Kronecker product of two matrices is a grid of blocks, each block a copy of B scaled by one entry of A .

The rule generalises: if $\vec{u} \in \mathbb{C}^m$ and $\vec{v} \in \mathbb{C}^n$, then $\vec{u} \otimes \vec{v} \in \mathbb{C}^{mn}$. **Dimensions multiply.** Two qubits need $2 \times 2 = 4$ numbers, three need 8, and n qubits need 2^n . That exponential growth is the reason a classical computer struggles to simulate even fifty qubits, and the reason you will write a NumPy simulator for small numbers of them in Assignment Part C.

WORKED EXAMPLE · A TENSOR PRODUCT OF TWO VECTORS

$$\begin{pmatrix} 1 \\ 2 \end{pmatrix} \otimes \begin{pmatrix} 3 \\ 4 \end{pmatrix} = \begin{pmatrix} 1 \cdot 3 \\ 1 \cdot 4 \\ 2 \cdot 3 \\ 2 \cdot 4 \end{pmatrix} = \begin{pmatrix} 3 \\ 4 \\ 6 \\ 8 \end{pmatrix}.$$

The order of the factors matters: $(3, 4) \otimes (1, 2) = (3, 6, 4, 8)$, a different vector.

Tensor product of two matrices

The same recipe works for matrices. If A is 2×2 and B is 2×2 ,

$$A \otimes B = \begin{pmatrix} A_{11}B & A_{12}B \\ A_{21}B & A_{22}B \end{pmatrix},$$

a 4×4 matrix made of four 2×2 blocks. In general an $m \times n$ matrix tensored with a $p \times q$ matrix is $mp \times nq$.

The key property links the two definitions:

$$(A \otimes B)(\vec{u} \otimes \vec{v}) = (A\vec{u}) \otimes (B\vec{v}).$$

In words: a matrix built as $A \otimes B$ acts on a product vector by letting A act on the first factor and B on the second, independently. In Unit 3 this is how you apply one gate to the first qubit and another to the second at the same time.

WORKED EXAMPLE · CHECKING THE PRODUCT RULE

Take $P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $Q = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$, $\vec{u} = (1, 2)$ and $\vec{v} = (3, 4)$.

$$P \otimes Q = \begin{pmatrix} 0 \cdot Q & 1 \cdot Q \\ 1 \cdot Q & 0 \cdot Q \end{pmatrix} = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}.$$

Left side. $\vec{u} \otimes \vec{v} = (3, 4, 6, 8)$, and multiplying by $P \otimes Q$ row by row gives $(6, -8, 3, -4)$.

Right side. $P\vec{u} = (2, 1)$ and $Q\vec{v} = (3, -4)$, so $P\vec{u} \otimes Q\vec{v} = (2 \cdot 3, 2 \cdot (-4), 1 \cdot 3, 1 \cdot (-4)) = (6, -8, 3, -4)$.

The two sides agree.

Not every vector is a tensor product

Every tensor product of two vectors in $\mathbb{C}^2$ has the form $(w_1, w_2, w_3, w_4) = (ac, ad, bc, bd)$. Multiply the outer pair and the inner pair:

$$w_1 w_4 = acbd = w_2 w_3.$$

So any vector in $\mathbb{C}^4$ with $w_1 w_4 \neq w_2 w_3$ **cannot** be written as $\vec{u} \otimes \vec{v}$, whatever $\vec{u}$ and $\vec{v}$ you try.

WORKED EXAMPLE · A VECTOR THAT DOES NOT SPLIT

Take $\vec{w} = \frac{1}{\sqrt{2}}(1, 0, 0, 1)$. Here $w_1w_4 = \frac{1}{2}$ and $w_2w_3 = 0$. These differ, so $\vec{w}$ is not a tensor product of two vectors.

You can also see it directly. $ad = 0$ forces $a = 0$ or $d = 0$. If $a = 0$ then $ac = 0$, but $w_1 \neq 0$. If $d = 0$ then $bd = 0$, but $w_4 \neq 0$. Neither works.

By contrast $\frac{1}{2}(1, 1, 1, 1) = \frac{1}{\sqrt{2}}(1, 1) \otimes \frac{1}{\sqrt{2}}(1, 1)$ passes the test: $\frac{1}{4} = \frac{1}{4}$.

A two-qubit state that cannot be split into a state for each qubit is called **entangled**, and the vector $\vec{w}$ above is the most famous example of one; it returns in Unit 3 as the Bell state.

NOTE · CHECK IT IN NUMPY

```
P = np.array([[0, 1], [1, 0]]); Q = np.array([[1, 0], [0, -1]])
u = np.array([1, 2]); v = np.array([3, 4])
print(np.kron(u, v))                # [3 4 6 8]
lhs = np.kron(P, Q) @ np.kron(u, v)
rhs = np.kron(P @ u, Q @ v)
print(lhs, rhs)                    # [ 6 -8  3 -4] twice
w = np.array([1, 0, 0, 1]) / np.sqrt(2)
print(w[0]*w[3], w[1]*w[2])        # 0.5 0.0: not a product
```

Summary

IDEA	FORMULA	WATCH OUT FOR
Vector in $\mathbb{C}^n$	column of n complex entries	entries numbered from 1
Scalar multiple	$c\vec{v}$ multiplies every entry by c	$i \cdot i = -1$
Dagger	$\vec{v}^\dagger = (\vec{v}^*)^T, (A^\dagger)_{jk} = A_{kj}^*$	conjugate and transpose
Matrix times vector	$(A\vec{v})_j = \sum_k A_{jk}v_k$	columns of A = rows of $\vec{v}$
Matrix product	$(AB)_{jk} = \sum_l A_{jl}B_{lk}$	$AB \neq BA$ in general
Dagger of a product	$(AB)^\dagger = B^\dagger A^\dagger$	order reverses
2×2 inverse	$\frac{1}{ad-bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$	none if $ad - bc = 0$
Inner product	$\langle \vec{u}, \vec{v} \rangle = \sum_k u_k^* v_k$	conjugate the first argument
Norm	$\ \vec{v}\ = \sqrt{\sum_k v_k ^2}$	$ 4i ^2 = 16$, not -16
Orthogonal	$\langle \vec{u}, \vec{v} \rangle = 0$	the conjugate matters
Orthonormal basis	$\langle \vec{e}_j, \vec{e}_k \rangle = 1$ if $j = k$, else 0	orthogonal is not enough
Expansion	$\vec{v} = \sum_k \langle \vec{e}_k, \vec{v} \rangle \vec{e}_k$	basis vector in the first slot
Pythagoras	$\ \vec{v}\ ^2 = \sum_k \langle \vec{e}_k, \vec{v} \rangle ^2$	the seed of the Born rule
Hermitian	$M^\dagger = M$	real diagonal, real eigenvalues
Unitary	$U^\dagger U = I$	preserves norms; columns orthonormal
Eigenvector	$A\vec{v} = \lambda\vec{v}, \vec{v} \neq \vec{0}$	defined only up to a factor
Characteristic equation	$\lambda^2 - (a+d)\lambda + (ad-bc) = 0$	roots may be complex
Tensor product	$(a, b) \otimes (c, d) = (ac, ad, bc, bd)$	dimensions multiply
Product rule	$(A \otimes B)(\vec{u} \otimes \vec{v}) = A\vec{u} \otimes B\vec{v}$	not every vector is a product

NUMPY	DOES
<code>np.array([1+2j, 3])</code>	a complex vector; <code>1j</code> is i
<code>A @ B</code> , <code>A @ v</code>	matrix products (never <code>*</code>)
<code>A.conj().T</code>	the dagger
<code>np.vdot(u, v)</code>	$\langle \vec{u}, \vec{v} \rangle$, conjugating the first argument
<code>np.linalg.norm(v)</code>	$\ \vec{v}\ $
<code>np.linalg.inv(A)</code>	A^{-1}
<code>np.linalg.eigh(M)</code>	eigenvalues and eigenvector columns of a Hermitian M
<code>np.kron(A, B)</code>	$A \otimes B$, for vectors or matrices

Self-assessment

Attempt every question by hand before you read the solutions. Use NumPy to check your arithmetic afterwards, not instead.

Vectors and matrices

1. Let $\vec{u} = (2, -i)$ and $\vec{v} = (1 + i, 3)$. Compute $\vec{u} + \vec{v}$, $(1 - i)\vec{u}$ and $2\vec{u} - i\vec{v}$.
2. Write down $\vec{v}^\dagger$ for $\vec{v} = (1 - i, 2i, 3)$, and $A^\dagger$ for $A = \begin{pmatrix} 1 & 2i \\ 3 - i & 4 \end{pmatrix}$.
3. Compute $A\vec{x}$ for $A = \begin{pmatrix} 1 & 2 \\ -1 & i \end{pmatrix}$ and $\vec{x} = (i, 1)$, once with rows and once as a combination of columns.
4. The matrix A is 2×3 , B is 3×2 , $\vec{v} \in \mathbb{C}^3$ and $\vec{w} \in \mathbb{C}^2$. Which of AB , BA , $A\vec{v}$, $A\vec{w}$, $B\vec{w}$ and $\vec{v}^T B$ are defined? Give the shape of each one that is.
5. Compute AB and BA for $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$.
6. Find the inverse of $\begin{pmatrix} 3 & 1 \\ 4 & 2 \end{pmatrix}$ and check it.
7. Explain why $\begin{pmatrix} 2 & 4 \\ 1 & 2 \end{pmatrix}$ has no inverse, and find a non-zero vector it sends to $\vec{0}$.

Inner products, norms and bases

8. Compute $\langle \vec{u}, \vec{v} \rangle$ and $\langle \vec{v}, \vec{u} \rangle$ for $\vec{u} = (1, i)$ and $\vec{v} = (i, 2 + i)$.
9. Find the norm of $\vec{v} = (1 + i, 1 - 2i)$ and normalise it.
10. Find the complex number x for which $(1 + i, 1)$ and $(1, x)$ are orthogonal.
11. Show that $\vec{g}_1 = \frac{1}{\sqrt{2}}(1, i)$ and $\vec{g}_2 = \frac{1}{\sqrt{2}}(1, -i)$ form an orthonormal basis of $\mathbb{C}^2$.
12. Find the component of $\vec{v} = (3, 4)$ along $\vec{e} = \frac{1}{\sqrt{2}}(1, 1)$, the projection vector, and check that the leftover is orthogonal to $\vec{e}$.
13. Expand $\vec{v} = (1, i)$ in the diagonal basis $\vec{f}_1 = \frac{1}{\sqrt{2}}(1, 1)$, $\vec{f}_2 = \frac{1}{\sqrt{2}}(1, -1)$. Verify that the expansion rebuilds $\vec{v}$ and that Pythagoras holds.
14. Expand $(0, 1)$ in the basis $\{\vec{g}_1, \vec{g}_2\}$ of question 11. What are $|c_1|^2$ and $|c_2|^2$?
15. A classmate using a mathematics textbook computes $\sum_k u_k v_k^*$ for the vectors of question 8. What will they get, and how is it related to your answer?

Special matrices, eigenvalues and tensor products

16. For each matrix, say whether it is symmetric, Hermitian, unitary: $A = \begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix}$, $B = \begin{pmatrix} 0 & i \\ -i & 0 \end{pmatrix}$,

$$C = \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix}, D = \begin{pmatrix} i & 0 \\ 0 & 1 \end{pmatrix}.$$

17. For which values of a and b is $\begin{pmatrix} a & 2+i \\ b & 5 \end{pmatrix}$ Hermitian?

18. Show that $U = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ i & -i \end{pmatrix}$ is unitary. What do you notice about its columns?

19. Find the eigenvalues and normalised eigenvectors of $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$.

20. Find the eigenvalues and eigenvectors of the Hermitian matrix $\begin{pmatrix} 1 & i \\ -i & 1 \end{pmatrix}$, and check that the eigenvectors are orthogonal.

21. Compute $(1, -1) \otimes (2, i)$ and $I_2 \otimes \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. How many entries does a vector describing three qubits have?

22. Which of these vectors in $\mathbb{C}^4$ are tensor products of two vectors in $\mathbb{C}^2$? (a) $\frac{1}{2}(1, 1, 1, 1)$, (b) $\frac{1}{\sqrt{2}}(0, 1, 1, 0)$, (c) $\frac{1}{2}(1, i, 1, i)$.

1. $\vec{u} + \vec{v} = (3 + i, 3 - i)$. $(1 - i)\vec{u} = (2 - 2i, (1 - i)(-i)) = (2 - 2i, -i + i^2) = (2 - 2i, -1 - i)$. $2\vec{u} - i\vec{v} = (4 - i(1 + i), -2i - 3i) = (4 - i + 1, -5i) = (5 - i, -5i)$.

2. $\vec{v}^\dagger = (1 + i \quad -2i \quad 3)$, a row. For A , conjugate every entry to get $\begin{pmatrix} 1 & -2i \\ 3 + i & 4 \end{pmatrix}$, then transpose: $A^\dagger = \begin{pmatrix} 1 & 3 + i \\ -2i & 4 \end{pmatrix}$.

3. Rows: $1 \cdot i + 2 \cdot 1 = 2 + i$ and $-1 \cdot i + i \cdot 1 = 0$. Columns: $i(1, -1) + 1(2, i) = (i + 2, -i + i) = (2 + i, 0)$. So $A\vec{x} = (2 + i, 0)$.

4. $AB: (2 \times 3)(3 \times 2)$, defined, 2×2 . $BA: (3 \times 2)(2 \times 3)$, defined, 3×3 . $A\vec{v}: (2 \times 3)(3 \times 1)$, defined, a vector in $\mathbb{C}^2$. $A\vec{w}: (2 \times 3)(2 \times 1)$, **not** defined. $B\vec{w}: (3 \times 2)(2 \times 1)$, defined, a vector in $\mathbb{C}^3$. $\vec{v}^T B: (1 \times 3)(3 \times 2)$, defined, a 1×2 row.

5. $AB = \begin{pmatrix} 1+1 & 0+1 \\ 0+1 & 0+1 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$ and $BA = \begin{pmatrix} 1 & 1 \\ 1 & 1+1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$. They differ, so these matrices do not commute.

6. $\det = 3 \cdot 2 - 1 \cdot 4 = 2$, so the inverse is $\frac{1}{2} \begin{pmatrix} 2 & -1 \\ -4 & 3 \end{pmatrix} = \begin{pmatrix} 1 & -1/2 \\ -2 & 3/2 \end{pmatrix}$. Check: row 1 of the matrix times column 1 of the inverse is $3 \cdot 1 + 1 \cdot (-2) = 1$; row 1 times column 2 is $3(-\frac{1}{2}) + 1 \cdot \frac{3}{2} = 0$; row 2 times column 1 is $4 - 4 = 0$; row 2 times column 2 is $-2 + 3 = 1$. The product is I .

7. $\det = 2 \cdot 2 - 4 \cdot 1 = 0$. The second column is twice the first, so the matrix sends the whole plane onto a line and cannot be undone. Using View 2, $2 \cdot (\text{column 1}) - 1 \cdot (\text{column 2}) = \vec{0}$, so $(2, -1)$ is sent to $\vec{0}$: $(2 \cdot 2 + 4 \cdot (-1), 1 \cdot 2 + 2 \cdot (-1)) = (0, 0)$.

8. $\langle \vec{u}, \vec{v} \rangle = 1^* \cdot i + i^*(2 + i) = i - i(2 + i) = i - 2i - i^2 = 1 - i$. $\langle \vec{v}, \vec{u} \rangle = i^* \cdot 1 + (2 + i)^* \cdot i = -i + (2 - i)i = -i + 2i + 1 = 1 + i$, the conjugate, as property 1 promises.

9. $|1 + i|^2 = 2$ and $|1 - 2i|^2 = 1 + 4 = 5$, so $\|\vec{v}\| = \sqrt{7}$ and $\hat{v} = \frac{1}{\sqrt{7}}(1 + i, 1 - 2i)$.

10. $\langle (1 + i, 1), (1, x) \rangle = (1 - i) \cdot 1 + 1 \cdot x = 0$, so $x = -1 + i$. (With the other order, $\langle (1, x), (1 + i, 1) \rangle = 1 + i + x^* = 0$ gives $x^* = -1 - i$, the same x .)

11. $\|\vec{g}_1\|^2 = \frac{1}{2}(|1|^2 + |i|^2) = 1$ and likewise $\|\vec{g}_2\|^2 = 1$. $\langle \vec{g}_1, \vec{g}_2 \rangle = \frac{1}{2}(1 \cdot 1 + (-i)(-i)) = \frac{1}{2}(1 + i^2) = 0$. Two orthonormal vectors in $\mathbb{C}^2$ are linearly independent (neither is a multiple of the other, since a multiple would have a non-zero inner product with it), and any two independent vectors in $\mathbb{C}^2$ form a basis.

12. $\langle \vec{e}, \vec{v} \rangle = \frac{1}{\sqrt{2}}(3 + 4) = \frac{7}{\sqrt{2}} \approx 4.95$. Projection: $\frac{7}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}}(1, 1) = (\frac{7}{2}, \frac{7}{2})$. Leftover: $(3, 4) - (3.5, 3.5) = (-0.5, 0.5)$, and $\langle \vec{e}, (-0.5, 0.5) \rangle = \frac{1}{\sqrt{2}}(-0.5 + 0.5) = 0$.

13. $c_1 = \langle \vec{f}_1, \vec{v} \rangle = \frac{1}{\sqrt{2}}(1 + i)$ and $c_2 = \langle \vec{f}_2, \vec{v} \rangle = \frac{1}{\sqrt{2}}(1 - i)$ (the entries of $\vec{f}_k$ are real, so conjugating them changes nothing). Rebuild: $\frac{1}{2}((1 + i)(1, 1) + (1 - i)(1, -1)) = \frac{1}{2}(2, 2i) = (1, i)$. Pythagoras: $|c_1|^2 = \frac{1}{2} \cdot 2 = 1$, $|c_2|^2 = 1$, total 2, and $\|\vec{v}\|^2 = 1 + 1 = 2$.

14. $c_1 = \langle \vec{g}_1, (0, 1) \rangle = \frac{1}{\sqrt{2}}(1 \cdot 0 + (-i) \cdot 1) = -\frac{i}{\sqrt{2}}$ and $c_2 = \langle \vec{g}_2, (0, 1) \rangle = \frac{1}{\sqrt{2}}(i) = \frac{i}{\sqrt{2}}$. Check: $-\frac{i}{\sqrt{2}}\vec{g}_1 + \frac{i}{\sqrt{2}}\vec{g}_2 = \frac{1}{2}((-i, 1) + (i, 1)) = (0, 1)$. $|c_1|^2 = |c_2|^2 = \frac{1}{2}$, which add to $1 = \|(0, 1)\|^2$.

15. $\sum_k u_k v_k^* = 1 \cdot (-i) + i(2 - i) = -i + 2i + 1 = 1 + i$. That is the complex conjugate of our $\langle \vec{u}, \vec{v} \rangle = 1 - i$, or equivalently our $\langle \vec{v}, \vec{u} \rangle$. The two conventions always differ by exactly this conjugation.

16. A : symmetric and (being real) Hermitian; not unitary, since its first column has norm $\sqrt{5}$. B : $B^\dagger = \begin{pmatrix} 0 & (-i)^* \\ i^* & 0 \end{pmatrix} = \begin{pmatrix} 0 & i \\ -i & 0 \end{pmatrix} = B$, so Hermitian; not symmetric ($i \neq -i$); its columns $(0, -i)$ and $(i, 0)$ are orthonormal, so also unitary. C : symmetric, but $C^\dagger = \begin{pmatrix} 1 & -i \\ -i & 1 \end{pmatrix} \neq C$, so not Hermitian; $C^\dagger C = 2I$, so not unitary (though $C/\sqrt{2}$ is). D : diagonal, hence symmetric; not Hermitian because the diagonal entry i is not real; unitary, because $D^\dagger D = \text{diag}(-i \cdot i, 1) = I$.

17. The diagonal must be real, so a can be any real number. The off-diagonal entries must be conjugates: $b = (2 + i)^* = 2 - i$.

18. $U^\dagger = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -i \\ 1 & i \end{pmatrix}$. Then $U^\dagger U = \frac{1}{2} \begin{pmatrix} 1 + (-i)(i) & 1 + (-i)(-i) \\ 1 + i \cdot i & 1 + i(-i) \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = I$. The columns are $\frac{1}{\sqrt{2}}(1, i) = \vec{g}_1$ and $\frac{1}{\sqrt{2}}(1, -i) = \vec{g}_2$: the complex orthonormal basis from question 11, as the column test predicts.

19. $a + d = 0$ and $ad - bc = -1$, so $\lambda^2 - 1 = 0$ and $\lambda = \pm 1$. For $\lambda = 1$: $-x + y = 0$, eigenvector $\frac{1}{\sqrt{2}}(1, 1)$. For $\lambda = -1$: $x + y = 0$, eigenvector $\frac{1}{\sqrt{2}}(1, -1)$. These are $\vec{f}_1$ and $\vec{f}_2$. The matrix is Hermitian and unitary, and its eigenvalues are both real and of modulus 1.

20. $a + d = 2$ and $ad - bc = 1 - i(-i) = 1 + i^2 = 0$, so $\lambda^2 - 2\lambda = 0$ and $\lambda = 0$ or 2 . For $\lambda = 2$: $-x + iy = 0$, so $x = iy$; with $y = -i$ this gives $(1, -i)$. Check: $(1 + i(-i), -i + (-i)) = (2, -2i) = 2(1, -i)$. For $\lambda = 0$: $x + iy = 0$; with $y = i$, $x = 1$, giving $(1, i)$. Check: $(1 + i \cdot i, -i + i) = (0, 0)$. Orthogonal: $\langle (1, -i), (1, i) \rangle = 1 + (i)(i) = 0$. Both eigenvalues are real, as they must be.

21. $(1, -1) \otimes (2, i) = (1 \cdot 2, 1 \cdot i, -1 \cdot 2, -1 \cdot i) = (2, i, -2, -i)$. With $P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $I_2 \otimes P =$

$$\begin{pmatrix} 1 \cdot P & 0 \cdot P \\ 0 \cdot P & 1 \cdot P \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}. \text{ Three qubits: } 2 \times 2 \times 2 = 8 \text{ entries.}$$

22. (a) $w_1 w_4 = \frac{1}{4} = w_2 w_3$; it is $\frac{1}{\sqrt{2}}(1, 1) \otimes \frac{1}{\sqrt{2}}(1, 1)$, a product. (b) $w_1 w_4 = 0$ but $w_2 w_3 = \frac{1}{2}$, so it is **not** a product (it is entangled). (c) $w_1 w_4 = \frac{1}{4} \cdot i$ and $w_2 w_3 = \frac{1}{4} \cdot i$; indeed $\frac{1}{2}(1, i, 1, i) = \frac{1}{\sqrt{2}}(1, 1) \otimes \frac{1}{\sqrt{2}}(1, i)$, a product.

Where to go next

Unit 3 is where the mathematics of this unit becomes physics. The only new thing at the start is notation: $\vec{v}$ from this unit is written $|v\rangle$ there, a “ket”, and $\vec{v}^\dagger$ is written $\langle v|$, a “bra”, so the inner product $\langle \vec{u}, \vec{v} \rangle$ becomes $\langle u|v \rangle$. After that, the four postulates of quantum mechanics use exactly what you have practised here: unit vectors for states, unitary matrices for operations, projections and Pythagoras for measurement probabilities, Hermitian matrices and their eigenvalues for observables, and the tensor product for several qubits.

Before session 10, make sure you can do questions 8 to 14 of the self-assessment without looking at the solutions. They are the calculations Unit 3 uses every week.

The class examination (multiple choice, on Evolveus) is held after week 9 and covers Units 1 and 2. **Assignment Part B** of the Qubit Workbook covers this unit and is due at the end of week 10; it asks for the same hand calculations as the self-assessment, each followed by a short NumPy check, so the snippets in these notes are a good template for your answers.